

Food Science B - Food Science Division B Test - University of Chicago B - 01-15-2022

The code word for the video is: cotton candy

Google form for video submission: <https://forms.gle/FhY5J4LyZAJ4Tc6T7> (<https://forms.gle/FhY5J4LyZAJ4Tc6T7>)

1. (1.00 pts) Which of the following incorrectly describes a disaccharide sugar?

- ☐ A) Sucrose is a combination of glucose and fructose.
- ☐ B) Lactose is a combination of glucose and galactose.
- ☐ C) Maltose is a combination of glucose and glucose with a 1,4 glycosidic bond.
- ☒ D) Chitobiose is a combination of glucose and glucose with a 1,1 glycosidic bond.

2. (1.00 pts)

Phenylketonuria (PKU) is characterized by the lack of enzyme ____, which allows for the breakdown of phenylalanine, an amino acid found in the sweetener ____.

- ☐ A) Phenylalanine dehydrogenase; advantame
- ☐ B) Phenylalanine dehydrogenase; aspartame
- ☒ C) Phenylalanine hydroxylase; aspartame
- ☐ D) Phenylalanine hydroxylase; advantame

3. (1.00 pts) Which of the following is an incorrect statement about canning?

- ☐ A) One downside of canning is a loss of nutrients during the process.
- ☒ B) Canning involves first heating and then packaging food in a vacuumed container.
- ☐ C) Canning is a type of food preservation.
- ☐ D) Parts of the food that are closer to the heat source are typically overheated.

4. (1.00 pts) Benedict's Test involves a reaction between a reducing sugar's ____ and the ____ in Benedict's solution.

- ☒ A) Aldehyde; copper sulfate
- ☐ B) Aldehyde; copper sulfide
- ☐ C) Ketone; copper sulfate
- ☐ D) Ketone; copper sulfide

5. (1.00 pts) The Maillard reaction occurs best at which temperature range?

- ☐ A) 160 to 185 °C
- ☐ B) 120 to 145 °C
- ☒ C) 140 to 165 °C
- ☐ D) 100 to 125 °C

6. (1.00 pts) Which of the following is an incorrect step of the Krebs cycle?

- ☐ A) Malate dehydrogenase turns malate into oxaloacetate
- ☐ B) Aconitase turns citrate into isocitrate
- ☐ C) Succinate dehydrogenase turns succinate into fumarate
- ☒ D) Isocitrate dehydrogenase turns isocitrate into succinyl-CoA

7. (1.00 pts) Which of the following is not a product of the Krebs cycle?

- ☐ A) CO₂
- ☒ B) O₂
- ☐ C) NADH
- ☐ D) FADH₂

8. (1.00 pts) In which of the sugar stages does caramelization occur?

- ☒ A) Brown liquid stage
- ☐ B) Hard crack stage
- ☐ C) Firm ball stage
- ☐ D) Burnt Liquid stage

9. (1.00 pts) Which of the following is a non-essential biological macromolecule?

- ☐ A) Omega-3 fatty acids
- ☐ B) Galactose
- ☒ C) Cysteine
- ☐ D) Isoleucine

10. (1.00 pts) Which class of enzymes is only found in the stomachs of ruminant mammals?

- ☐ A) Amylase
- ☐ B) Oxireductase
- ☒ C) Rennet
- ☐ D) Protease

11. (1.00 pts) What is the most commonly used font in nutrition labels?

- ☒ A) Helvetica
- ☐ B) Calibri
- ☐ C) Arial
- ☐ D) Times New Roman

12. (1.00 pts) Which allergen is currently not required to be labeled?

- ☐ A) Shellfish
- ☐ B) Soybeans

- ☐ C) Tree nuts
- ☒ D) Sesame

13. (1.00 pts) Which disease is characterized by a gluten allergy?

- ☐ A) Peptic ulcer disease
- ☒ B) Celiac disease
- ☐ C) Crohn's disease
- ☐ D) Irritable bowel syndrome

14. (1.00 pts) Which of the following is false regarding leavening agents?

- ☐ A) These include anything that can produce a gas that will lighten and expand dough or batter.
- ☒ B) Salt is a leavening agent, as it controls the rate of yeast fermentation, which impacts how much dough can rise.
- ☐ C) During leavening, gas gets trapped in a matrix of gluten and starch.
- ☐ D) Baking soda is a good leavening agent, but only works well in the presence of moisture and an acidic substance.

15. (1.00 pts)

An acidified food (AF) is a low-acid food to which acid(s) or acid food(s) are added and which has a finished equilibrium pH of ____ or below and a water activity (aw) greater than ____.

- ☒ A) 4.6; 0.85
- ☐ B) 4.6; 0.75
- ☐ C) 3.6; 0.75
- ☐ D) 3.6; 0.95

16. (1.00 pts) Which of the following is the correct definition of water activity?

- ☐ A) The difference between the partial vapor pressure of water in a food substance to the partial vapor pressure of distilled water at the same temperature.
- ☐ B) The ratio of the partial vapor pressure of distilled water to the partial vapor pressure of water in a food substance at the same temperature.
- ☒ C) The ratio of the partial vapor pressure of water in a food substance to the partial vapor pressure of distilled water at the same temperature.
- ☐ D) The ratio of the partial vapor pressure of water in a food substance to the partial vapor pressure of deionized water at the same temperature.

17. (1.00 pts) Which of the following is an enzyme that breaks down proteins into amino acids and is usually found in the small intestines?

- ☐ A) Kinase
- ☐ B) Chymosin
- ☐ C) Pepsin
- ☒ D) Trypsin

18. (1.00 pts) Which of the following is true regarding the brown bag test?

- ☐ A) If the bag is more transparent, that means there are less proteins
- ☒ B) If the bag is less transparent, that means there are less lipids
- ☐ C) If the bag is more transparent, that means there are less lipids
- ☐ D) If the bag is less transparent, that means there are more proteins

19. (1.00 pts) Which of the following is not in Biuret's solution?

- ☐ A) Copper(II) sulfate
- ☐ B) Sodium hydroxide
- ☒ C) Sodium citrate
- ☐ D) Potassium sodium tartrate

20. (1.00 pts) Which temperature is in the "Danger Zone" for food?

- ☐ A) 138° C
- ☒ B) 138° F
- ☐ C) 32° F
- ☐ D) 220° F

21. (2.00 pts) Name the 6 high-intensity sweeteners that are currently approved by the FDA

Expected Answer: Saccharin, Aspartame, Acesulfame potassium (Ace-K), Sucralose, Neotame, Advantame

22. (2.00 pts) Describe the reaction and/or complex formation that occurs in Biuret's test.

Expected Answer: Formation of complex between the cupric ions in copper sulfate and the lone pair of electrons present on the nitrogen and oxygen atoms of peptide bonds of proteins

23. (2.50 pts) What do each of the following acronyms stand for?

- a) NAD+
- b) FAD
- c) ADP
- d) GTP
- e) CoA

Expected Answer: a) Nicotinamide adenine dinucleotide b) flavin adenine dinucleotide c) Adenosine diphosphate d) Guanosine triphosphate e) Coenzyme A

24. (5.00 pts) List at least FIVE changes that occurred to nutrition labels in January of 2021.

Expected Answer: Here are a few answers: The new nutrition labels increase the font size of the number of calories and remove the number of calories from fat, along with the serving size now being on the bottom of the number of servings per container, bolded and formatted differently as well. The percentage of daily values were also updated according to the new serving sizes of foods and beverages. Other changes include showing amounts of vitamins and minerals, replacing vitamins A and C with D and potassium since Americans don't get enough of these two, but do more of A and C, adding the amount of added sugar, and replacing the old footnote with a new one. Please reference here for more possible answers: <https://www.gravelshea.com/2020/06/changes-to-nutrition-facts-label-coming-january-2021/>

25. (8.00 pts) Label the various enzymes A-H in the diagram below.

Expected Answer: A) Citrate synthase B) Aconitase C) Isocitrate dehydrogenase D) Alpha-ketoglutarate dehydrogenase E) succinyl-CoA synthetase F) Succinate dehydrogenase G) Fumarase H) Malate dehydrogenase

26. (3.00 pts)

When regular and diet soda are both put in water, regular soda sinks and diet soda floats. However, a packet of regular mayonnaise floats higher than a packet of light mayonnaise. Explain both of these phenomena.

Expected Answer: Answers must involve a discussion of DENSITY. Sugar is more dense than water, so more sugar (regular cola) means more sinking. Conversely, fat is less dense than water, so more fat (regular mayonnaise) means more floating.

27. (3.00 pts)

Lyophilization is a method of drying foods for the purpose of food preservation. Describe the process of lyophilization. Specifically, distinguish between primary and secondary drying.

Expected Answer: Lyophilization is essentially freeze drying food for preservation purposes. It involves putting frozen food under a vacuum and allowing it to dry via low pressure vacuum (sublimation). Primary drying describes the sublimation stage - very low temperature combined with vacuum-like pressures. Secondary drying occurs afterwards when there are still some "unfreezable" water molecules left due to bonds between the water and dried substance. These are dried by bringing the temperature up a little bit and then again applying a vacuum to cause evaporation/sublimation.

28. (2.00 pts) Describe the difference between nucleation and crystallization.

Expected Answer: The big thing here is understanding that nucleation is the first step of crystallization, which describes the overall process of solid formation in cold liquid solutions.

Use the food label below to answer questions 29-32:

29. (1.00 pts) What are “added sugars”?

Expected Answer: Added sugars are sweeteners added to food. They do not provide any nutritional value.

30. (2.00 pts) How many mg of potassium are in 100g of this food? Show units.

Expected Answer: 25.6mg

31. (2.00 pts) What is the structural difference between saturated and trans fats? Which one is less healthy?

Expected Answer: Saturated fats have a cis linkage, whereas trans have a trans linkage (it's in the name). Trans fats are considered less healthy than saturated fats.

32. (2.00 pts) Based on this nutrition label, how many mg of calcium are in one container of this food if the recommended daily value for calcium is 260mg? Show units.

Expected Answer: 416mg

33. (2.00 pts) What is the difference between moisture content and water activity. Which one is more useful for the purposes of food preservation, and why?

Expected Answer: Moisture content is simply the amount of water in a food, but water activity describes the amount of “free water” as a ratio of partial pressures. Therefore, water activity is more useful, as it tells us how much microorganisms may interact with a certain food substance.

34. (2.00 pts) List all of the fat-soluble essential vitamins.

Expected Answer: Vitamins A, D, E, K. Take off points for any other answers.

35. (8.00 pts) Name the sugar stage, describe the properties of sugar, and give the percentage of sugar in water at the following temperatures.

- a) 237° F
- b) 305° F
- c) 338° F
- d) 354° F

Expected Answer: a) Soft ball stage; forms a soft flexible ball in cold water; 85% b) Hard crack stage; after being in water, if you try to bend it, it breaks; 95% c) Brown liquid stage; caramelization occurs and sugar turns brown; 100% d) Burnt liquid stage; highest stage where liquid is dark and no longer candy; 100%

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Congrats on finishing, and thank you for participating! Best of luck with your other events :)